Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2026-27 (ODD)**	**Course Name**	**HPCA Lab**
Semester	**5**	**Course Code**	**23ADS1524**
Roll No	**119**	**Name of Student**	**SHRUSTI KATAKWAR**

Practical Number	**05**
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	**Write a basic CUDA program (e.g., vector addition or image thresholding).**
Problem Definition	**Write a basic CUDA program (e.g., vector addition or image thresholding).**
Theory (100 words)	1. CUDA (Compute Unified Device Architecture) is a parallel computing platform developed by NVIDIA. 2. It enables programs to execute tasks on a Graphics Processing Unit (GPU). 3. CUDA improves performance by processing multiple data elements simultaneously. 4. Vector addition is a simple example of parallel computation. 5. In vector addition, two input vectors are added element-wise to produce an output vector. 6. CUDA programs use kernels, which are functions executed on the GPU. 7. Threads execute kernels, while blocks and grids organize threads. 8. Memory management involves allocating and transferring data between CPU and GPU. 9. The CPU launches the kernel and controls execution. 10. CUDA reduces computation time for large datasets.

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Procedure and Execution (100 Words)	Step for Implementation: 1. Install the NVIDIA CUDA Toolkit and configure the required environment. 2. Include the necessary CUDA header files in the program. 3. Initialize two input vectors with sample data. 4. Allocate memory on the GPU for input and output vectors. 5. Copy input data from CPU memory to GPU memory. 6. Write a CUDA kernel to perform element-wise vector addition. 7. Configure the number of blocks and threads for execution. 8. Launch the CUDA kernel on the GPU. 9. Copy the result from GPU memory back to CPU memory. 10. Display the output and verify the results. 11. Free allocated GPU memory after execution.

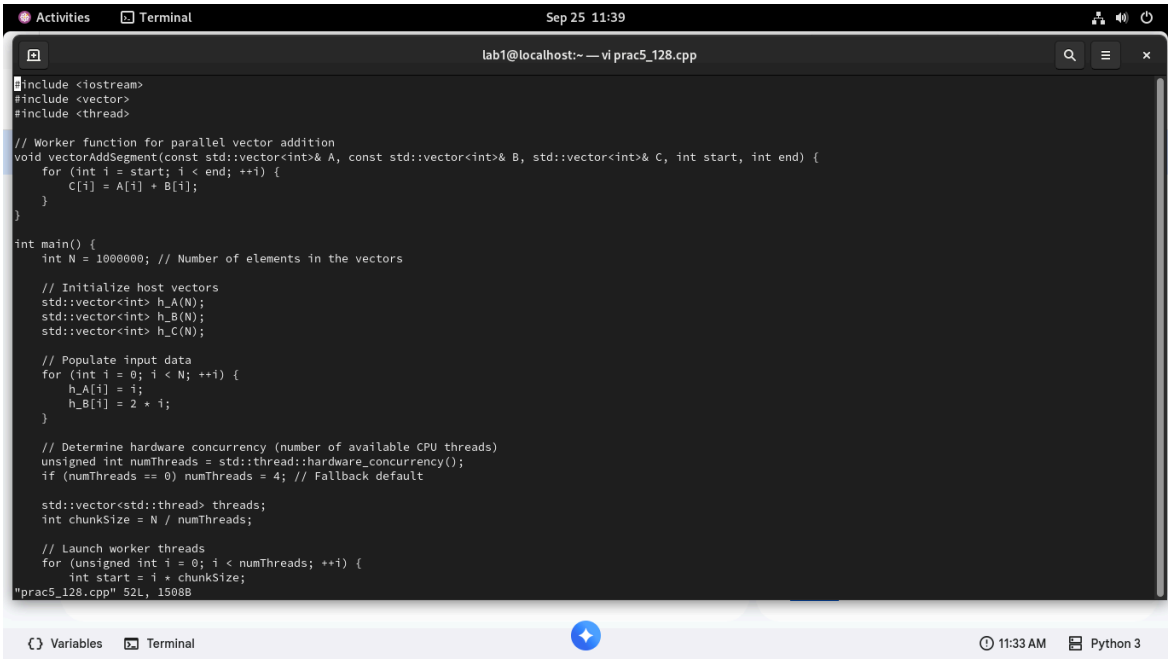
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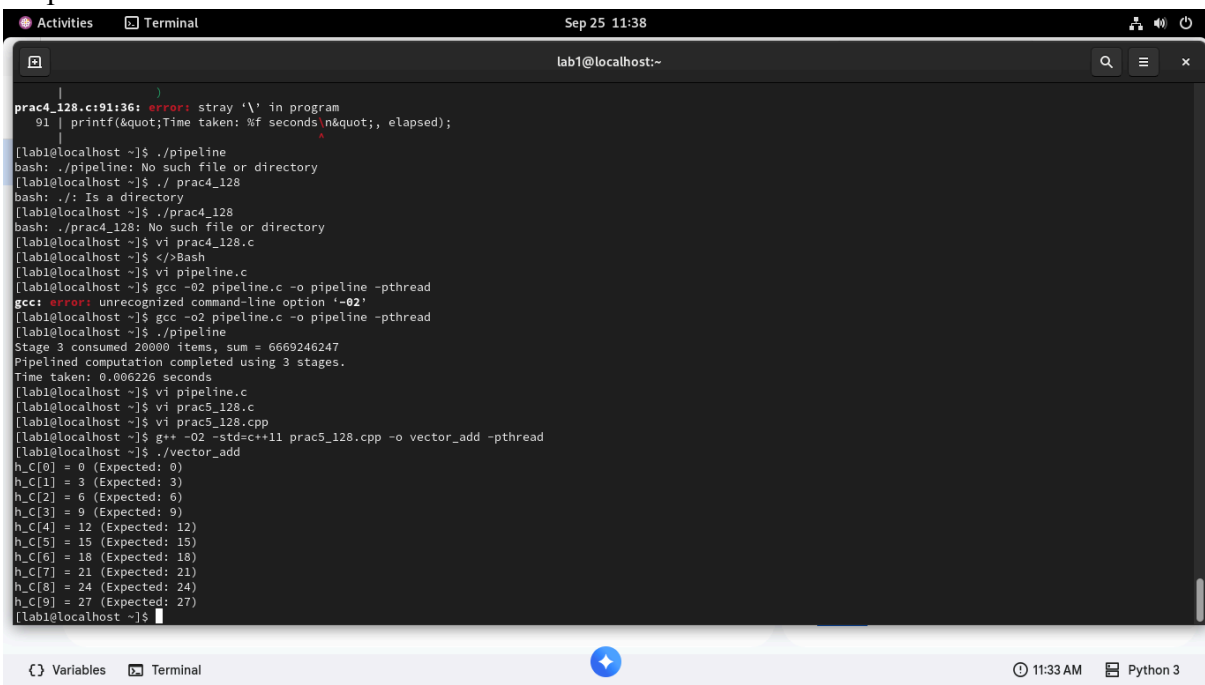
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Code:



Output:



Output Analysis	1. The CUDA program performs vector addition by adding corresponding elements of two input vectors. 2. Each GPU thread processes one or more vector elements.

	3. The computed results are stored in the output vector. 4. The output is transferred from GPU memory to CPU memory for display. 5. The results are verified by comparing them with the expected values. 6. Parallel execution improves performance for large datasets by utilizing multiple GPU threads. 7. The execution time depends on GPU capability, data size, and memory transfer overhead. 8. Thus, the program demonstrates the basic working of CUDA and the benefits of parallel computing.
Link of student Github profile where lab assignment has been uploaded	**https://github.com/Shrusti-Katakwar/HCPA**
Conclusion	1. The basic CUDA program demonstrates the implementation of parallel computing using a GPU. 2. Vector addition is performed efficiently by executing multiple threads simultaneously. 3. The program illustrates memory allocation, data transfer, kernel execution, and result retrieval. 4. CUDA reduces computation time for large datasets by utilizing GPU parallelism. 5. Thus, CUDA provides an effective approach for accelerating computationally intensive applications.

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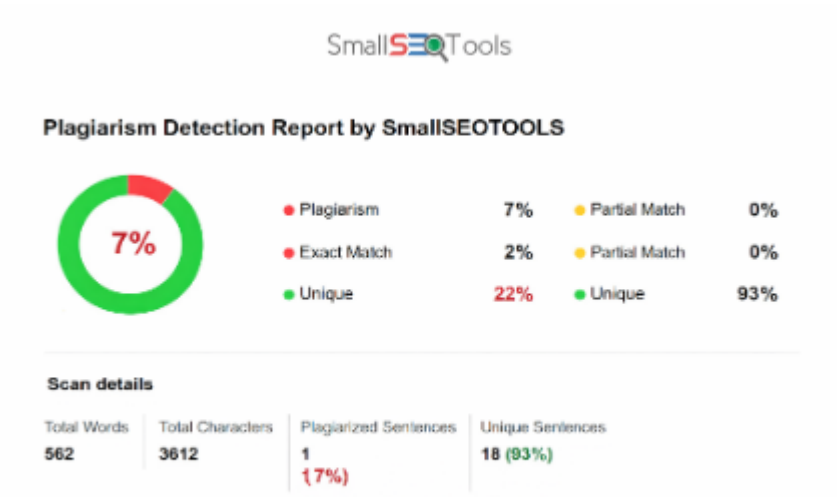
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Date	**10/10/2026**